FREIE UNIVERSITÄT BERLIN

# AI Speech and language variation & change

A study on manifestation of AI speech in human natural language

class: [16838] germanische sprachen im vergleich
tutor: Hüning / Konvicna
term: WS25/26

snc: 2026-02-25 11:34:49.740211

## Preliminary

inspired by the above paper which indeed found (evidence) for GPT influenced human language after the introduction of chatGPT we tried to replicate the pipeline of building an AI vocabulary (model preferred lemmata) and compare frequencies of model typical words across pre- and post model introduction human language corpora. The first draft essai proves their hypothesis that LLM generated language manifests within human natural language.

## Questions / Hypotheses

- do humans adapt to language produced by LLM and incorporate AI speech vocabulary into their own language production?
- H1: we will find higher frequencies of LLM typical vocabulary within human natural language corpora after onset of model introduction.

## p2

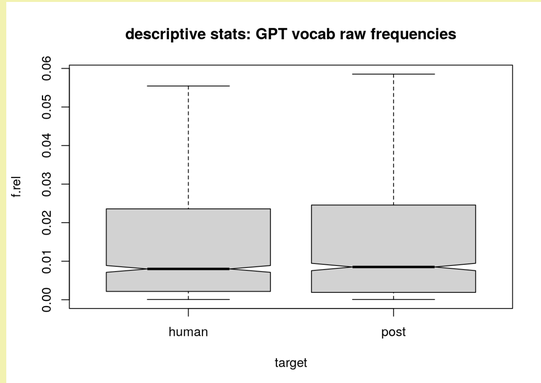


Figure 1: stats descriptive

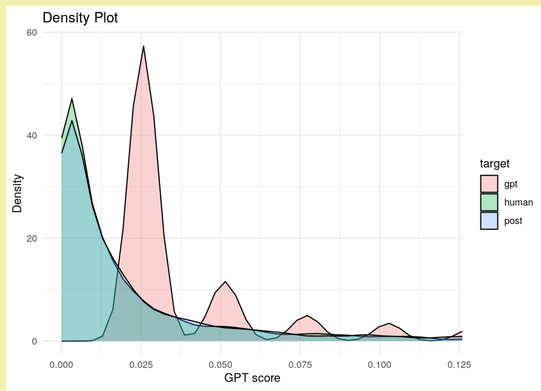


Figure 2: model vocab density

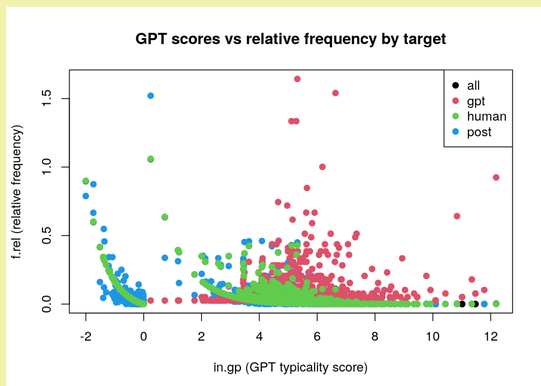


Figure 3: model vocab distribution

## Methods / Data

our human langugae data consists of raw texts from german bundestag plenary protocols (). the LLM corpus consists of model summaries of a first subset of these texts generated with the prompt you see left.

we first devised AI-typical lemmata in the model corpus which are distinct for that corpus using a linear regression model that calculates a score for each lemma in the corpus, see Figure 3 and Figure 2.

## Results

to gather an insight, yet with simple descriptive stats comparing the raw frequencies of gpt-preferred lemmas in pre- and post-gemini onset we find that in the target corpus the occurences of these lemma increase, see Figure 1.

## References


Gilquin (2008) Mehl (2021) UCSB Santa Barbara Corpus


```
"/SPUND

 fj$WOR
/plenar

$vcov@f
get)),

izer =
uman"

T))
h3[[1]]<-sh.H1[2]
  tdb6$f.norm_g<-mean(tdb6$f.rel[tdb6$target=="gpt"])/t
# ?unnest_tokens
qdf$value[m]<-d.end
#?boxplot

uare va                                                p < pt
  target<-unique(tdb6$target)

  encode = "json"
##########################################################
  return(ll)
f1p1<-data.frame(f1p,target="post",size=sum(f1p$FREQ))
fj9<-join.freqs(data.frame(fj8[,c(1,2)]),data.frame(fj7[,c(1,3)]))
# chisq.test(x)$p.value           # 0.4233
# head(fs,20)
eval(e)
# 20250125(15.03)
lran3$lemma<-rownames(lran3)
qdf
  theme_minimal()
     main = "GPT scores vs relative frequency by target"
  # tdb6$f.norm_all<-mean(tdb6$FREQ)/tdb6$FREQ
fs4[fs4$fa==0,]
# ### wks., joined frequencies of target=gpt+target=all, score for gpt
print(q)
p1
d.end<-range[2]
  mode(dfa$FREQ)<-"double"
r2
sum(m)
tokens<-get.tokens()
#
p.df$target[p.df$target=="0-human"]<-"human"
range<-c("2025-04-01","2025-05-30")

response <- POST(geturl, body = toJSON(request_body, auto_unbox = TRUE), encode = "json",add_headers(.headers=h3))
#   filter(str_detect(word, "^[A-Za-zäöüÄÖÜ]+$")) %>%  # Optional: alphabetic words only
sp<-sum(lmdf.c$freq[mp])
tab <-
    ungroup()
                                                       REQ))

                                                       df.c$target))),
# ti<-ti[order(ti,decreasing = T)]
     col  = as.factor(lmdf.c$target),
  )

head(fs2,10)
write_json(gg.json,paste0(Sys.getenv("GIT_TOP"),"/SPUND-LX/germanic/HA/textprompt.json"))
```